

A Short-Term Traffic Flow Prediction Method Based on SVR

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Abstract—Nowadays, the issue of urban traffic congestion is becoming more and more serious. Real-time and accurate short-term traffic flow forecasting is quite crucial in the urban intelligent transportation system, which helps traffic managers formulate relevant policies to alleviate traffic congestion and other issues. In this paper, a support vector regression model is established for the short-term traffic flow prediction, and the traffic flow in Xuancheng, Anhui from August 1 to August 7 is predicted. The experiments results show that the model proposed in this paper is superior to other models based on the RMSE, MSE and MAPE three evaluation indexes. The research in this paper improves the allocation and utilization efficiency of traffic resources.

Keywords—Traffic Flow Prediction; SVR; Short-Term; Intelligent Traffic Decision; traffic policies;

I. INTRODUCTION

With the rapid development of traffic, the rail transit network is often over-saturated. The accurate prediction of traffic flow during different periods has gradually become a research hotspot. It can not only provide the basis for traffic managers to formulate control measures, but also provide guidance information for traffic participants, so as to make traffic flow reasonably distributed, improve road traffic efficiency, reduce carbon emissions and alleviate road traffic congestion[1]. With the outbreak of information technology revolution and the development of big data and deep learning, it is possible to accurately predict traffic flow.

Traffic flow forecasting can be divided into short-term, medium-term and long-term traffic flow forecasting according to the forecasting interval length. Short-term traffic flow prediction is generally less than 15min. Short-term traffic flow prediction can predict the short-term traffic flow in the future and can better meet the real-time requirements of urban traffic, so it is often used for traffic flow prediction of urban traffic network. However, at the same time, short-term traffic flow also has strong time-varying and random characteristics, which increases the difficulty of forecasting.

Short-term traffic flow forecasting is one of the core issues in urban traffic research, so it has always been a hot spot in the field of traffic flow forecasting. So far, scholars have put forward a variety of short-term traffic flow forecasting models. The traffic flow prediction methods are also mainly divided into mathematical statistical models and machine learning methods. Traditional mathematical statistical models for short-term traffic flow prediction mainly include autoregressive integrated moving average model(ARIMA), Kalman filtering

model and grey model and so on. The machine learning methods mainly include neural networks, support vector regression (SVR), Long Short Term Memory Network(LSTM), etc. And the combination model based on the above two types is also applied to the prediction of short-term traffic flow. The above methods promoted the development of short-term prediction to a certain extent. For example, time series-based models are often used in traffic flow forecasting, and ARIMA is one of them. Zhao Peng et al. used ARIMA model to predict the passenger flow of Beijing subway in a short time. Research shows that the prediction error of this model is only 4%, and it has high prediction accuracy[2]. However, this model has higher requirements for time series data. Later, Chen Xueping established a short-term traffic flow prediction model by using BP neural network, and verified the feasibility of the model by using the traffic flow data of South China Expressway[3]. Although neural network has strong nonlinear fitting ability, it also has some shortcomings, such as the need for a large number of data sets and the determination of model parameters. Combining mathematical statistical models and machine learning methods, Sun Xianghai used the combination model of neural network and SARIMA to forecast short-term traffic flow, and the model showed good prediction performance in the experiment[4].

The traffic flow of a given traffic section is affected by many factors and has strong instability, which belongs to a typical nonlinear time series prediction problem[5][6]. SVR has achieved satisfactory results in predicting high-dimensional data with small samples and dealing with nonlinear problems and local minima. SVR is a regression algorithm developed on the basis of SVM. It is mainly used to solve regression problems, and its thinking is similar to that of SVM. SVR is widely used in transportation field because of its low structural risk, avoiding dimension disaster and over-fitting. Especially, Support vector machine has strong advantages in real-time traffic flow prediction, and is suitable for forecasting short-term traffic flow. Therefore, this paper chooses the support vector regression algorithm to predict short-term traffic flow.

In this paper, we obtained and process the lane-level coil data of the main road in Xuancheng City, Anhui Province in China. Next, we conduct classification training according to the date flow pattern within a week and constructed a prediction model based on training data. Next, we test the constructed short-term prediction model based on SVR model. Finally, we predict the traffic situation of the main road during the period to be predicted from August 1 to 7, and compare with the other three forecasting methods. The short-term traffic flow prediction model based on SVR can get much better results

than other algorithms on small sample training set. The experimental results also confirm this point. The reason why support vector machine has become one of the most commonly used and best classifiers at present lies in its excellent generalization ability, because its optimization goal is to minimize the structural risk, not the empirical risk. This is the innovation of this paper. Accurately predicting the traffic flow in different periods is helpful to alleviate traffic congestion and reduce traffic accidents, and also plays an important role in urban traffic road planning.

II. DATA AND METHODS

A. Data description and preprocessing

The data set used in this paper is the lane-level coil data of all main roads in Xuancheng, Anhui, from July 1st to July 31st at specific time periods, including 7: 00 a.m. to 9: 00 a.m., 11 a.m. to 1 p.m., 5: 00 p.m. to 7: 00 p.m., 8: 00 p.m. to 10: 00 p.m. The virtual measurement platform provides us with the data items, which includes DEVICEID, FTNODE, FROMTIME, TOTIME, INTERVAL, LANR_ID, LANENUM, COUNT and FLOW, etc. Based on our research objectives, three data items, namely LANEID, FROMTIME, TOTIME and FLOW, are selected for analysis. Taking July 8, 2019 as an example, some selected data are shown in Table I.

TABLE I. PARTIAL DATA PRESENTION ON JULY 8, 2019

LANEID	FROMTIME	TOTIME	FLOW/hour
1	6:00 a.m.	6:05 a.m.	12
2	6:00 a.m.	6:05 a.m.	36
3	6:00 a.m.	6:05 a.m.	18
1	6:05 a.m.	6:10 a.m.	12
...	...	...	...
2	9:55 p.m.	10:00 p.m.	36
3	9:55 p.m.	10:00 p.m.	48

To facilitate subsequent modeling and analysis, we first preprocess the data by the following steps:

Step 1. The processing of abnormal and missing data. For the abnormal data, we delete the data items with LANEID of -1. And for the missing data, we replace the missing data with the mean value of the data at two intervals before and after the missing data point.

Step 2. Dividing the traffic patterns and the training set and test set. Since different date in a week have different traffic patterns, we divide the traffic patterns and forecast each traffic pattern separately. For example, the traffic flow pattern is that people drive to work on weekdays. Another traffic flow pattern is that on weekends, the traffic flow of going out to play accounts for a certain proportion of the traffic flow, which is obviously different from the former traffic flow pattern. The periods of peak and trough in different traffic patterns are different. Therefore, it is necessary to divide the traffic patterns. In the paper, we divide the traffic patterns in a week into seven

types, which correspond to each day in a week. For the lane-level coil data of all main roads, it is divided into training set and test set which is according to the ratio of 4:1.

B. Establishment of SVR model

Support Vector Machine is a learning algorithm with sparse nonlinear kernel, which transforms the linear indivisible problem into a linear problem in high-dimensional space by kernel function, improves the generalization ability of the model by seeking the minimum structural risk, minimizes the empirical risk and confidence range, and seeks a compromise between the data fitting accuracy and the complexity of the approximation function to avoid over-fitting[7]. Its training principle is as follows [5].

Step 1. Mapping the input traffic flow training data to a high-dimensional feature space by using a nonlinear mapping function;

Step 2. The learning goal of SVR can be expressed by the following formula (1).

$$\begin{aligned} \min_{\omega, b, \xi_i, \xi_i^*} & \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \\ s.t. & \begin{cases} f(x)_i - y_i \leq \varepsilon + \xi_i \\ y_i - f(x)_i \leq \varepsilon + \xi_i^* \\ \xi_i \geq 0, \xi_i^* \geq 0, i = 1, 2, \dots, n. \end{cases} \end{aligned} \quad (1)$$

where, ξ_i is the lower relaxation variable and ξ_i^* is the upper relaxation variable, which corresponds to the insensitive range $y - f(x) \leq \varepsilon$. The first term $1/2 \|\omega\|^2$ of the formula (1) is the programming term, and the minimum programming term can improve the generalization ability of the model. The second term $C \sum_{i=1}^n (\xi_i + \xi_i^*)$ is the empirical error term. Super parameter

C is a regularization constant, which determines the trade-off between structural risk and empirical risk.

Step 3. Lagrange multiplier is introduced to obtain Lagrange function, which is then transformed into convex quadratic programming problem. Finally, the regression test result can be expressed by the following formula (2). And $K(x_i, x)$ stands for kernel function.

$$\hat{f}(x) = \sum_{i=1}^n (\alpha_i^* - \alpha_i) K(x_i, x) + b. \quad (2)$$

Step 4. In this paper, the sections of lanes 4819_4689, 4662_4689, 4697_4694, 4706_4697, 4635_4651 are respectively set and predicted. Traffic flow forecasting is a typical nonlinear time series forecasting problem. Through kernel function, traffic flow data in low-dimensional space can be mapped to high-dimensional space, which can be transformed into linear regression. In this paper, Gaussian radial basis function (RBF) kernel function is used, as shown in formula (3). And σ is the kernel parameter, and the value of it has an important impact on the performance of SVR model.

$$K(x_i, x) = \exp\left(-\frac{\|x_i - x\|^2}{\sigma^2}\right) \quad (3)$$

Based on the above formula, it can be seen that the configuration of penalty parameter C , insensitivity loss parameter ε and kernel parameter σ determines the learning efficiency and generalization ability of the model.

In this paper, the RFB kernel function kernel = 'rbf' is selected, and the empirical formula of parameters σ is $\sigma = k^{-1}$ [4], k is the number of attributes in the input data. In this paper, the specific period from Monday to Sunday and the corresponding traffic volume are taken as input values, and the total number of attributes is 2, so $\sigma = 0.5$. Besides, we can know $\gamma = 2$. For penalty factors C , larger penalty factors can improve the accuracy of prediction results, but too large penalty factors will cause over-learning state, thus affecting the accuracy of the final test sample set. According to expert experience, we set $C = 40$ and the loss function $\varepsilon = 0.1$. When forecasting different time periods and different lanes, these three parameters should be properly adjusted according to the actual situation.

III. RESULTS OF EXPERIMENT

The short-term traffic flow model is trained by Python language, and the model is tested on the test data set. From August 1 to August 7, the traffic data of the next hour is predicted by the historical traffic flow of the previous hour, and the prediction time interval is 5 minutes. Based on the training set and test set obtained by data preprocessing in proportion, we analyze the data of every day and every lane on a weekly basis, train and test one by one, and get their own fitting results. According to the date between one week, we will divide seven traffic patterns. Taking Monday as an example, we take 5 minutes as an interval from 5:00p.m. to 7:00p.m., and count 120 cross-section converted hourly flow training data and 30 cross-section converted hourly flow test data, and analyze the training set and test set, as shown in Figure I and Figure II. After calculation, it can be seen that the short-term traffic flow prediction model based on SVR has a good fitting degree on the training set, and the fitting degree of the training set is 0.72. On the test set, the fitting degree of our model is 0.46. Although the fitting degree of the model in the test set is lower than that in the training set, it is acceptable to a certain extent.

A. Estimating index

RMSE is the square root of the mean square error, which is sensitive to outliers. The greater the error, the greater the impact on RMSE. MAE is mainly used to calculate the absolute average error of prediction model. MAPE reflects the relative error of prediction value, which is used to measure the accuracy of prediction method. We compared the predicted values given by the model with the real values from August 1 to 7, and calculated the values of RMSE, MAE and MAPE three evaluation indexes respectively. The score table of the optimal model prediction results is shown in Table II.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \quad (4)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i| \quad (5)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right| \quad (6)$$

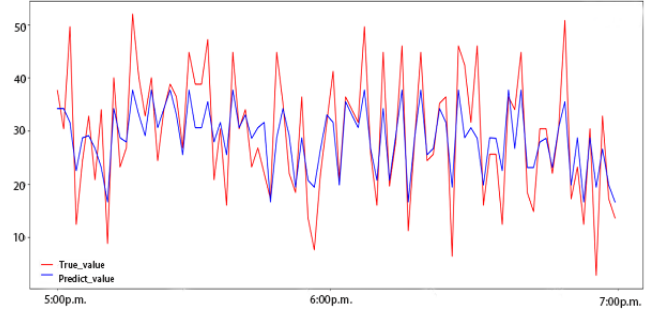


Figure I. Distribution of actual and fitted values of training set from 5:00p.m. to 7:00p.m. on Monday

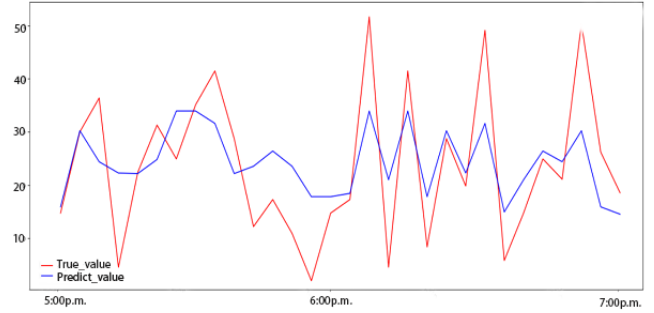


Figure II. Distribution of actual value and fitted value of test set from 5:00p.m. to 7:00p.m. on Monday

At present, the short-term prediction methods of traffic road flow mainly include ARIMA model, Kalman filtering model, artificial intelligence model, such as neural network, Support Vector Machine. Among them, ARIMA model is a classic model for traffic flow prediction. At the same time, some scholars also use BP neural network to predict the traffic flow, and achieved good results. In order to better illustrate the reliability of the experiment, this paper compares SVR model with ARIMA model, LSTM model and BP neural network model. In this paper, we set the autoregressive term in ARIMA model to 3, the difference order to 1, and the moving average term coefficient to 1. In this paper, we use two LSTM layers and two fully connected layers to construct LSTM circulating neural network. The training times of BP neural network are 54 times, and the learning rate is initially set to 0.01. After many experiments, it is found that 0.006 is the best prediction result. Table II shows the comparison of the evaluation indexes of these three prediction models. It can be seen from the table that the three error evaluation indexes of SVR are lower than those of BP neural network. Besides, the evaluation

index RMSE of SVR model is reduced by 14.3% compared with BP neural network. At the same time, the training time of BP neural network is longer than that of SVR model, which is also a disadvantage of BP neural network in predicting short-term traffic flow. Although the RMAE of ARIMA model and LSTM model is lower than that of SVR model, the other two indexes MAE and MAPE are higher than that of SVR model. Therefore, in this paper, to a certain extent, SVR model is superior to ARIMA model and LSTM model. And, the comparison of the three forecasting models proves the effectiveness of SVR model in short-term traffic flow forecasting.

TABLE II. COMPARISON OF EVALUATION INDEXES OF THREE FORECASTING MODELS

<i>model</i>	<i>RMSE</i>	<i>MAE</i>	<i>MAPE</i>
SVR	14.97	9.38	0.64
ARIMA	13.38	11.23	0.79
LSTM	14.45	11.76	0.81
BP	17.47	12.85	0.74

IV. CONCLUSION

In order to predict the traffic flow from August 1st to 7th, this paper establishes a short-term traffic flow prediction model based on SVR, and evaluates the prediction results by using three indexes. At the same time, this research compares the forecasting model with the other three forecasting models, and proves that SVR is superior to other methods. The model has excellent generalization ability, avoids some problems caused by over-fitting, can accurately predict urban short-term

traffic flow, and help urban managers take timely response measures to road sections that may be congested in the future, improving urban traffic problems and the utilization efficiency of traffic resources to a certain extent.

However, since this paper forecasts the future short-term traffic flow just based on historical data, and does not consider other factors, such as weather, holidays and traffic accidents, resulting in a large deviation between the predicted flow and the actual flow. In future research, other scholars can comprehensively predict short-term traffic flow by taking other factors into account.

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